

ADVANCE MACHINE LEARNING

QUIZ 3

Name: _____ Reg No: _____

CLO 2: Apply supervised, unsupervised, learning techniques to solve classification and clustering problems of moderate complexity.

Q No. 1: A software company wants to predict whether an employee will Leave (L) or Stay (S) based on the following attributes:

ID	Attrition	Experience	Job Satisfaction	Monthly Overtime	Performance Rating
1	L	Low	Low	Yes	Average
2	L	Low	Medium	Yes	Good
3	S	Medium	High	No	Good
4	S	High	Medium	No	Excellent
5	L	Medium	Low	Yes	Average
6	S	High	High	No	Excellent
7	L	Low	Medium	Yes	Average
8	S	Medium	Medium	No	Good
9	L	High	Low	Yes	Good
10	S	Medium	High	No	Excellent
11	L	Low	Low	No	Average
12	S	High	Medium	Yes	Excellent

PART A:

1. Compute the entropy of the full dataset.
2. Calculate the Information Gain for all attributes:
 - a. Experience
 - b. Job Satisfaction
 - c. Monthly Overtime
 - d. Performance Rating
3. Select the best root node and justify your choice mathematically.
4. Construct the decision tree.

PART B:

Using your constructed tree, classify the following employees:

- Employee X = (Medium, Low, Yes, Good)
- Employee Y = (High, High, No, Excellent)
- Employee Z = (Low, Medium, No, Average)

For each classification, show the decision path from root to leaf.